

Cinema Tickets System Design

1. High-Level Overview of Functions and Use Cases

A Cinema Ticket Management System handles all aspects of booking, scheduling, and viewing movies at a cinema complex.

Function	Description	Use Case Example
Movie Management	Adding, updating, and removing movies, including details like title, duration, genre, and rating.	A cinema administrator adds a new blockbuster movie set to be released next week.
Screen/Theater Management	Defining the physical screens (auditoriums), their capacity, and seat layout.	The system stores that "Screen 3" has 150 standard seats and 10 VIP seats.
Showtime Scheduling	Creating specific time slots for movies to be shown on a particular screen.	A user checks the available times for <i>Dune: Part Two</i> on Friday at 5:00 PM and 8:00 PM.
Ticket Booking/Sales	Allowing customers to select a movie, showtime, and specific seats, then process the payment.	A customer uses the website to book two seats for the latest action movie.
Payment Processing	Handling various payment methods (credit card, gift	The system securely processes a credit card payment for a booking.

	card, etc.) and generating a unique transaction ID.	
Seat Management	Keeping track of which seats are available, reserved, or sold for a given showtime.	A seat changes from "available" to "sold" immediately after a booking is confirmed.
User/Customer Management	Creating accounts for customers, storing their basic information, and tracking their booking history.	A registered user logs in and views all their past ticket purchases.
Reporting	Generating reports on sales, popular movies, and screen utilization for management.	The cinema manager pulls a report showing total revenue generated by the top 5 movies this month.

2. Entities, Attributes, and Relationships

This section details the core components of the system, their descriptive data, and how they connect.

A. Entities and Attributes:

Entity	Primary Key (PK)	Key Attributes	Data Type Examples
Movie	<u>MovieID</u>	Title, Duration (min), Genre, Rating, ReleaseDate	INT, VARCHAR(255), INT, VARCHAR(50),

			VARCHAR(10), DATE
Screen	ScreenID	Name (e.g., "Screen 1"), Capacity, ScreenType (e.g., 3D, IMAX)	INT, VARCHAR(50), INT, VARCHAR(50)
Showtime	ShowtimeID	ShowtimeDateTime, TicketPrice, MovieID (FK), ScreenID (FK)	INT, DATETIME, DECIMAL(5, 2), INT, INT
Customer	CustomerID	FirstName, LastName, Email, PhoneNumber	INT, VARCHAR(100), VARCHAR(100), VARCHAR(255), VARCHAR(15)
Booking	BookingID	BookingDateTime, TotalAmount, PaymentStatus, CustomerID (FK)	INT, DATETIME, DECIMAL(6, 2), VARCHAR(20), INT
Seat	SeatID	SeatNumber, RowNumber, SeatType (e.g., Standard, VIP), ScreenID (FK)	INT, VARCHAR(10), VARCHAR(10), VARCHAR(20), INT
Ticket	TicketID	BookingID (FK), ShowtimeID (FK), SeatID (FK), PricePaid	INT, INT, INT, INT, DECIMAL(5, 2)

B. Relationships

Relationship	Entities Involved	Cardinality	Description
Shows	Movie and Showtime	One-to-Many (1:M)	A Movie can have many Showtime instances, but each Showtime is for only one Movie .
Hosts	Screen and Showtime	One-to-Many (1:M)	A Screen can host many Showtime instances, but each Showtime is held in only one Screen .
Has Seats	Screen and Seat	One-to-Many (1:M)	A Screen has many Seats , but each Seat belongs to only one Screen .
Made By	Customer and Booking	One-to-Many (1:M)	A Customer can make many Bookings , but each Booking is made by only one Customer .
Includes	Booking and Ticket	One-to-Many (1:M)	A Booking can include many Tickets , but each Ticket belongs to only one Booking .
Is For	Showtime and Ticket	One-to-Many (1:M)	A Showtime can be associated with many Tickets , and each Ticket is for a specific Showtime .

Occupies	Seat and Ticket	One-to-One (1:1)	For a <i>specific showtime and booking</i> , a Ticket corresponds to exactly one Seat (ensuring no double-booking).
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3. ER Diagram

